QPM Model

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% =====================================
% Basic Quarterly Projection Model (QPM)
% =====================================

!transition_variables
'Real GDP (100*log)'                                            L_GDP
'Trend in Real GDP (100*log)'                                   L_GDP_BAR
'Output Gap (in %)'                                             L_GDP_GAP
'Quarterly Growth in Real GDP(in % pa)'                         DLA_GDP
'Real GDP Growth YoY (in % pa)'                                 D4L_GDP
'Real GDP Trend Growth QoQ annualized (in % pa)'                DLA_GDP_BAR
'Real NON-MINING GDP (100*log)'                                            L_GDP_OTH
'Trend in Real NON-MINING GDP (100*log)'                                   L_GDP_OTH_BAR
'NON-MINING Output Gap (in %)'                                             L_GDP_OTH_GAP
'Quarterly Growth in Real NON-MINING GDP(in % pa)'                         DLA_GDP_OTH
'Real NON-MINING GDP Growth YoY (in % pa)'                                 D4L_GDP_OTH
'Real NON-MINING GDP Trend Growth QoQ annualized (in % pa)'                DLA_GDP_OTH_BAR
'Real MINING GDP (100*log)'                                            L_GDP_MINE
'Trend in Real MINING GDP (100*log)'                                   L_GDP_MINE_BAR
'MINING Output Gap (in %)'                                             L_GDP_MINE_GAP
'Quarterly Growth in Real MINING GDP(in % pa)'                         DLA_GDP_MINE
'Real MINING GDP Growth YoY (in % pa)'                                 D4L_GDP_MINE
'Real MINING GDP Trend Growth QoQ annualized (in % pa)'                DLA_GDP_MINE_BAR

'Real YEARLY GDP (100*log)'   L_GDP_yearly
'Real GDP Growth Yearly (in % pa)'   DL_GDP_yearly
'Real MINING YEARLY GDP (100*log)'   L_GDP_MINE_yearly
'Real MINING GDP Growth Yearly (in % pa)'   DL_GDP_MINE_yearly
'Real NON-MINING YEARLY GDP (100*log)'   L_GDP_OTH_yearly
'Real NON-MINING GDP Growth Yearly (in % pa)'   DL_GDP_OTH_yearly


'Real Monetary Condition Index (in % pa)'                       MCI

'CPI (level, 100*log)'                                          L_CPI
'CPI Inflation QoQ annualized (in % pa)'                        DLA_CPI
'Expected CPI Inflation QoQ annualized (in % pa)'               E_DLA_CPI
'Expected CPI Inflation YoY (in % pa)'                   E_D4L_CPI
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'CPI Inflation YoY (in % pa)'                                            D4L_CPI
'Inflation Target (in % pa)'                                             D4L_CPI_TAR

'Real Marginal Cost  (in %)'                                             RMC

'Nominal Exchange Rate (LCY/FCY, 100*log)'                               L_S
'Nominal Exch. Rate Depreciation QoQ annualized (in % pa)'               DLA_S
'Nominal Exch. Rate Depreciation YoY (in % pa)'                          D4L_S
'Country Risk Premium (in % pa)'                                         PREM

'Nominal Policy Interest Rate (in % pa)'                                 RS
'Real Interest Rate (in % pa)'                                           RR
'Trend Real Interest Rate (in % pa)'                                     RR_BAR
'Real Interest Rate Gap (in %)'                                          RR_GAP
'Nominal Policy Neutral Interest Rate (in % pa)'                         RSNEUTRAL

'Real Exchange Rate (level, 100*log)'                                    L_Z
'Trend Real Exchange Rate (level, 100*log)'                              L_Z_BAR
'Real Exchange Rate Gap (in %)'                                          L_Z_GAP
'Real Exchange Rate Depreciation QoQ annualized (in % pa)'               DLA_Z
'Trend Real Exchange Rate Depreciation QoQ annualized(in % pa)'          DLA_Z_BAR

'Foreign Output Gap (in %)'                                              L_GDP_RW_GAP
'Foreign Nominal Interest Rate (in % pa)'                                RS_RW
'Foreign Real Interest Rate (in % pa)'                                   RR_RW
'Foreign Real Interest Rate Trend (in % pa)'                             RR_RW_BAR
'Foreign Real Interest Rate Gap (in %)'                                  RR_RW_GAP
'Foreign CPI (level, 100*log)'                                           L_CPI_RW
'Foreign Inflation QoQ annualized (in % pa)'                             DLA_CPI_RW

%% TUNES
TUNE_L_GDP_GAP
TUNE_L_GDP_OTH_GAP
TUNE_L_GDP_MINE_GAP
TUNE_SHK_L_GDP_OTH_GAP
TUNE_SHK_DLA_GDP_OTH_BAR
TUNE_SHK_RS
TUNE_SHK_DLA_CPI


% ------------------------------------------------------------------------------ %
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!transition_shocks
'Shock: Output gap (demand)'                                    SHK_L_GDP_GAP⟨σ=0.01⟩
'Shock: Mining output gap (demand)'                                    SHK_L_GDP_MINE_GAP⟨σ=2⟩
'Shock: Nonmining output gap (demand)'                                    SHK_L_GDP_OTH_GAP⟨σ=2⟩
'Shock: CPI inflation (cost-push)'                              SHK_DLA_CPI⟨σ=0.75⟩
'Shock: Exchange rate (UIP)'                                    SHK_L_S⟨σ=1.6⟩
'Shock: Interest rate (monetary policy)'                        SHK_RS⟨σ=1⟩
'Shock: Inflation target'                                       SHK_D4L_CPI_TAR⟨σ=2⟩

'Shock: Real interest rate'                                     SHK_RR_BAR⟨σ=0.5⟩
'Shock: Real exchange rate depreciation'                        SHK_DLA_Z_BAR⟨σ=0.4⟩
%'Shock: Potential GDP growth'                                  SHK_DLA_GDP_BAR
'Shock: Mining potential GDP growth'                            SHK_DLA_GDP_MINE_BAR⟨σ=1⟩
'Shock: Nonmining potential GDP growth'                         SHK_DLA_GDP_OTH_BAR⟨σ=1⟩

'Shock: Foreign output gap'                                     SHK_L_GDP_RW_GAP⟨σ=1⟩
'Shock: Foreign nominal interest rate'                          SHK_RS_RW⟨σ=1⟩
'Shock: Foreign inflation'                                      SHK_DLA_CPI_RW⟨σ=2⟩
'Shock: Foreign real interest rate'                             SHK_RR_RW_BAR⟨σ=0.5⟩

% ---------------------------------------------------------------------------- %
!parameters
%%b1 b2 b3 b4
b1_mine⟨0.7⟩ b3_mine⟨0.5⟩
b1_oth⟨0.7⟩ b2_oth⟨0.2⟩ b3_oth⟨0⟩
w_mine⟨0.15⟩
b4⟨0.4⟩
a1⟨0.6⟩ a2⟨0.3⟩ a3⟨0.5⟩
e1⟨0.4⟩
g1⟨0.8⟩ g2⟨1⟩ g3⟨0.1⟩

rho_D4L_CPI_TAR⟨0.8⟩
rho_DLA_Z_BAR⟨0.8⟩
rho_RR_BAR⟨0.8⟩
rho_DLA_GDP_BAR⟨0.8⟩
rho_DLA_GDP_MINE_BAR⟨0.8⟩
rho_DLA_GDP_OTH_BAR⟨0.8⟩

rho_L_GDP_RW_GAP⟨0.8⟩
rho_RS_RW⟨0.8⟩
rho_DLA_CPI_RW⟨0.8⟩
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```
rho_RR_RW_BAR⟨0.8⟩
ss_D4L_CPI_TAR⟨7⟩
ss_DLA_Z_BAR⟨−1⟩
ss_RR_BAR⟨3⟩
ss_DLA_GDP_BAR⟨NaN⟩
ss_DLA_GDP_MINE_BAR⟨6⟩
ss_DLA_GDP_OTH_BAR⟨5⟩
ss_DLA_CPI_RW⟨3⟩
ss_RR_RW_BAR⟨1.35⟩
%% --------------------------------------------------------------------------- %
!transition_equations
%% === Aggregate demand (the IS curve) ===
%%L_GDP_GAP = b1*L_GDP_GAP{-1} - b2*MCI + b3*L_GDP_RW_GAP + SHK_L_GDP_GAP;
L_GDP_MINE_GAP = b1_mine⟨0.7⟩*L_GDP_MINE_GAP{-1} + b3_mine⟨0.5⟩*L_GDP_RW_GAP + SHK_L_GDP_MINE_GAP⟨σ=2⟩;
L_GDP_OTH_GAP = b1_oth⟨0.7⟩*L_GDP_OTH_GAP{-1} - b2_oth⟨0.2⟩*MCI + b3_oth⟨0⟩*L_GDP_RW_GAP + SHK_L_GDP_OTH_GAP⟨σ=2⟩;
L_GDP_GAP = w_mine⟨0.15⟩*L_GDP_MINE_GAP + (1-w_mine⟨0.15⟩)*L_GDP_OTH_GAP + SHK_L_GDP_GAP⟨σ=0.01⟩;
%-- Real monetary conditions index
MCI = b4⟨0.4⟩*RR_GAP + (1-b4⟨0.4⟩)*(- L_Z_GAP);
%% === Inflation (the Phillips curve) ===
DLA_CPI =  a1⟨0.6⟩*DLA_CPI{-1} + (1-a1⟨0.6⟩)*DLA_CPI{+1} + a2⟨0.3⟩*RMC + SHK_DLA_CPI⟨σ=0.75⟩;
%-- Real marginal cost
RMC = a3⟨0.5⟩*L_GDP_GAP + (1-a3⟨0.5⟩)*L_Z_GAP;
%- expected inflation
E_DLA_CPI = DLA_CPI{+1};
E_D4L_CPI = D4L_CPI{+4};
%% === Monetary policy reaction function (a forward-looking Taylor-type Rule) ===
RS = g1⟨0.8⟩*RS{-1} + (1-g1⟨0.8⟩)*(RSNEUTRAL + g2⟨1⟩*(D4L_CPI{+4} - D4L_CPI_TAR{+4}) + g3⟨0.1⟩*L_GDP_GAP) + SHK_RS⟨σ=1⟩;
%- Neutral nominal policy interest rate
RSNEUTRAL = RR_BAR + D4L_CPI{+1};
%% === Modified Uncovered Interest Rate Parity (UIP) condition ===
L_S = (1-e1⟨0.4⟩)*L_S{+1} + e1⟨0.4⟩*(L_S{-1} + 2/4*(D4L_CPI_TAR - ss_DLA_CPI_RW⟨3⟩ + DLA_Z_BAR)) + (- RS + RS_RW + PREM)/4 + SHK_L_S⟨σ
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%% === Definitions ===

%- Fisher equation (RIR)
RR = RS - D4L_CPI{+1};

%- Real exchange rate (RER)
L_Z = L_S + L_CPI_RW - L_CPI;

%- Long-term version of UIP (consistency of trends)
DLA_Z_BAR{+1} = RR_BAR - RR_RW_BAR - PREM;

%% === Identities ===
L_GDP_yearly      = (L_GDP + L_GDP{-1} + L_GDP{-2} + L_GDP{-3})/4;
L_GDP_MINE_yearly = (L_GDP_MINE + L_GDP_MINE{-1} + L_GDP_MINE{-2} + L_GDP_MINE{-3})/4;
L_GDP_OTH_yearly  = (L_GDP_OTH + L_GDP_OTH{-1} + L_GDP_OTH{-2} + L_GDP_OTH{-3})/4;

DL_GDP_yearly     = L_GDP_yearly - L_GDP_yearly{-4};
DL_GDP_MINE_yearly     = L_GDP_MINE_yearly - L_GDP_MINE_yearly{-4};
DL_GDP_OTH_yearly     = L_GDP_OTH_yearly - L_GDP_OTH_yearly{-4};


DLA_GDP_BAR = 4*(L_GDP_BAR - L_GDP_BAR{-1});
DLA_GDP_MINE_BAR = 4*(L_GDP_MINE_BAR - L_GDP_MINE_BAR{-1});
DLA_GDP_OTH_BAR = 4*(L_GDP_OTH_BAR - L_GDP_OTH_BAR{-1});
DLA_Z_BAR   = 4*(L_Z_BAR - L_Z_BAR{-1});
DLA_Z       = 4*(L_Z - L_Z{-1});
DLA_GDP     = 4*(L_GDP - L_GDP{-1});
DLA_GDP_MINE     = 4*(L_GDP_MINE - L_GDP_MINE{-1});
DLA_GDP_OTH     = 4*(L_GDP_OTH - L_GDP_OTH{-1});
DLA_CPI     = 4*(L_CPI - L_CPI{-1});
DLA_S       = 4*(L_S - L_S{-1});

D4L_GDP     = L_GDP - L_GDP{-4};
D4L_GDP_MINE     = L_GDP_MINE - L_GDP_MINE{-4};
D4L_GDP_OTH     = L_GDP_OTH - L_GDP_OTH{-4};
D4L_CPI     = L_CPI - L_CPI{-4};
D4L_S       = L_S - L_S{-4};

%% === Gaps ===
RR_GAP    = RR - RR_BAR;
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L_Z_GAP   = L_Z - L_Z_BAR;
L_GDP_GAP = L_GDP - L_GDP_BAR;
L_GDP_MINE_GAP = L_GDP_MINE - L_GDP_MINE_BAR;
L_GDP_OTH_GAP = L_GDP_OTH - L_GDP_OTH_BAR;


%% === Trends ===
D4L_CPI_TAR = rho_D4L_CPI_TAR⟨0.8⟩*D4L_CPI_TAR{-1} + (1-rho_D4L_CPI_TAR⟨0.8⟩)*ss_D4L_CPI_TAR⟨7⟩ + SHK_D4L_CPI_TAR⟨σ=2⟩;
DLA_Z_BAR   = rho_DLA_Z_BAR⟨0.8⟩*DLA_Z_BAR{-1} + (1-rho_DLA_Z_BAR⟨0.8⟩)*ss_DLA_Z_BAR⟨−1⟩ + SHK_DLA_Z_BAR⟨σ=0.4⟩;
RR_BAR      = rho_RR_BAR⟨0.8⟩*RR_BAR{-1} + (1-rho_RR_BAR⟨0.8⟩)*ss_RR_BAR⟨3⟩ + SHK_RR_BAR⟨σ=0.5⟩;
% DLA_GDP_BAR = rho_DLA_GDP_BAR*DLA_GDP_BAR{-1} + (1-rho_DLA_GDP_BAR)*ss_DLA_GDP_BAR + SHK_DLA_GDP_BAR;
DLA_GDP_BAR = w_mine⟨0.15⟩*DLA_GDP_MINE_BAR + (1-w_mine⟨0.15⟩)*DLA_GDP_OTH_BAR;
DLA_GDP_MINE_BAR = rho_DLA_GDP_MINE_BAR⟨0.8⟩*DLA_GDP_MINE_BAR{-1} + (1-rho_DLA_GDP_MINE_BAR⟨0.8⟩)*ss_DLA_GDP_MINE_BAR⟨6⟩ + SHK_DLA_GDP
DLA_GDP_OTH_BAR = rho_DLA_GDP_OTH_BAR⟨0.8⟩*DLA_GDP_OTH_BAR{-1} + (1-rho_DLA_GDP_OTH_BAR⟨0.8⟩)*ss_DLA_GDP_OTH_BAR⟨5⟩ + SHK_DLA_GDP_OTH_

%% === Foreign Sector Equations ===
L_GDP_RW_GAP = rho_L_GDP_RW_GAP⟨0.8⟩*L_GDP_RW_GAP{-1} + SHK_L_GDP_RW_GAP⟨σ=1⟩;
DLA_CPI_RW   = rho_DLA_CPI_RW⟨0.8⟩*DLA_CPI_RW{-1} + (1-rho_DLA_CPI_RW⟨0.8⟩)*ss_DLA_CPI_RW⟨3⟩ + SHK_DLA_CPI_RW⟨σ=2⟩;
RS_RW        = rho_RS_RW⟨0.8⟩*RS_RW{-1} + (1-rho_RS_RW⟨0.8⟩)*(RR_RW_BAR + DLA_CPI_RW) + SHK_RS_RW⟨σ=1⟩;
RR_RW_BAR    = rho_RR_RW_BAR⟨0.8⟩*RR_RW_BAR{-1} + (1-rho_RR_RW_BAR⟨0.8⟩)*ss_RR_RW_BAR⟨1.35⟩ + SHK_RR_RW_BAR⟨σ=0.5⟩;
RR_RW        = RS_RW - DLA_CPI_RW;
RR_RW_GAP    = RR_RW - RR_RW_BAR;

DLA_CPI_RW   = 4*(L_CPI_RW - L_CPI_RW{-1});

%% === Tune Identities ===
TUNE_L_GDP_GAP = L_GDP_GAP;
TUNE_L_GDP_OTH_GAP = L_GDP_OTH_GAP;
TUNE_L_GDP_MINE_GAP = L_GDP_MINE_GAP;
TUNE_SHK_L_GDP_OTH_GAP = SHK_L_GDP_OTH_GAP⟨σ=2⟩;
TUNE_SHK_DLA_GDP_OTH_BAR = SHK_DLA_GDP_OTH_BAR⟨σ=1⟩;
TUNE_SHK_RS = SHK_RS⟨σ=1⟩;
TUNE_SHK_DLA_CPI = SHK_DLA_CPI⟨σ=0.75⟩;

%% ------------------------------------------------------------------------------ %
!measurement_variables
OBS_L_GDP
OBS_L_GDP_MINE
OBS_L_GDP_OTH
OBS_L_CPI
OBS_RS
```

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OBS_L_S
OBS_D4L_CPI_TAR

OBS_L_GDP_RW_GAP
OBS_DLA_CPI_RW
OBS_RS_RW
OBS_L_GDP_GAP
OBS_L_GDP_OTH_GAP
OBS_L_GDP_MINE_GAP
OBS_TUNE_SHK_L_GDP_OTH_GAP
OBS_TUNE_SHK_DLA_GDP_OTH_BAR
OBS_L_Z_GAP
OBS_TUNE_SHK_RS
OBS_TUNE_SHK_DLA_CPI

!measurement_equations
OBS_L_GDP = L_GDP;
OBS_L_GDP_MINE = L_GDP_MINE;
OBS_L_GDP_OTH = L_GDP_OTH;
OBS_L_CPI = L_CPI;
OBS_RS    = RS;
OBS_L_S   = L_S;
OBS_D4L_CPI_TAR = D4L_CPI_TAR;

OBS_L_GDP_RW_GAP = L_GDP_RW_GAP;
OBS_DLA_CPI_RW   = DLA_CPI_RW;
OBS_RS_RW        = RS_RW;

OBS_L_Z_GAP = L_Z_GAP;

%% tune variables
OBS_L_GDP_GAP = TUNE_L_GDP_GAP;
OBS_L_GDP_OTH_GAP = TUNE_L_GDP_OTH_GAP;
OBS_L_GDP_MINE_GAP = TUNE_L_GDP_MINE_GAP;
OBS_TUNE_SHK_L_GDP_OTH_GAP = TUNE_SHK_L_GDP_OTH_GAP;
OBS_TUNE_SHK_DLA_GDP_OTH_BAR =  TUNE_SHK_DLA_GDP_OTH_BAR;
OBS_TUNE_SHK_RS = TUNE_SHK_RS;
OBS_TUNE_SHK_DLA_CPI = TUNE_SHK_DLA_CPI;


%% ------------------------------------------------------------------------------- %
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Legend
_GAP        cyclical deviation from a trend
_BAR        trend (equilibrium)
ss_         steady-state value
DLA_        q-o-q change
D4L_        y-o-y change
_RW         foreign variable
SHK_        equation residual
```

Steady state

Variable	Description	Value
L_GDP_GAP	Output Gap (in %)	0
DLA_GDP	Quarterly Growth in Real GDP(in % pa)	5.15
D4L_GDP	Real GDP Growth YoY (in % pa)	5.15
DLA_GDP_BAR	Real GDP Trend Growth QoQ annualized (in % pa)	5.15
L_GDP_OTH_GAP	NON-MINING Output Gap (in %)	0
DLA_GDP_OTH	Quarterly Growth in Real NON-MINING GDP(in % pa)	5
D4L_GDP_OTH	Real NON-MINING GDP Growth YoY (in % pa)	5
DLA_GDP_OTH_BAR	Real NON-MINING GDP Trend Growth QoQ annualized (in % pa)	5
L_GDP_MINE_GAP	MINING Output Gap (in %)	0
DLA_GDP_MINE	Quarterly Growth in Real MINING GDP(in % pa)	6
D4L_GDP_MINE	Real MINING GDP Growth YoY (in % pa)	6
DLA_GDP_MINE_BAR	Real MINING GDP Trend Growth QoQ annualized (in % pa)	6
DL_GDP_yearly	Real GDP Growth Yearly (in % pa)	5.15
DL_GDP_MINE_yearly	Real MINING GDP Growth Yearly (in % pa)	6
DL_GDP_OTH_yearly	Real NON-MINING GDP Growth Yearly (in % pa)	5
MCI	Real Monetary Condition Index (in % pa)	0
DLA_CPI	CPI Inflation QoQ annualized (in % pa)	7
E_DLA_CPI	Expected CPI Inflation QoQ annualized (in % pa)	7
E_D4L_CPI	Expected CPI Inflation YoQ (in % pa)	7
D4L_CPI	CPI Inflation YoY (in % pa)	7
D4L_CPI_TAR	Inflation Target (in % pa)	7
RMC	Real Marginal Cost (in %)	0
DLA_S	Nominal Exch. Rate Depreciation QoQ annualized (in % pa)	3
D4L_S	Nominal Exch. Rate Depreciation YoY (in % pa)	3
PREM	Country Risk Premium (in % pa)	2.65
RS	Nominal Policy Interest Rate (in % pa)	10
RR	Real Interest Rate (in % pa)	3
RR_BAR	Trend Real Interest Rate (in % pa)	3
RR_GAP	Real Interest Rate Gap (in %)	0
RSNEUTRAL	Nominal Policy Neutral Interest Rate (in % pa)	10
L_Z_GAP	Real Exchange Rate Gap (in %)	0
DLA_Z	Real Exchange Rate Depreciation QoQ annualized (in % pa)	-1
DLA_Z_BAR	Trend Real Exchange Rate Depreciation QoQ annualized(in % pa)	-1

Variable	Description	Value
L_GDP_RW_GAP	Foreign Output Gap (in %)	0
RS_RW	Foreign Nominal Interest Rate (in % pa)	4.35
RR_RW	Foreign Real Interest Rate (in % pa)	1.35
RR_RW_BAR	Foreign Real Interest Rate Trend (in % pa)	1.35
RR_RW_GAP	Foreign Real Interest Rate Gap (in %)	0
DLA_CPI_RW	Foreign Inflation QoQ annualized (in % pa)	3
TUNE_L_GDP_GAP		0
TUNE_L_GDP_OTH_GAP		0
TUNE_L_GDP_MINE_GAP		0
TUNE_SHK_L_GDP_OTH_GAP		0
TUNE_SHK_DLA_GDP_OTH_BAR		0
TUNE_SHK_RS		0
TUNE_SHK_DLA_CPI		0